

Self-Supervised Multi-Modal World Model *with* 4D Space-Time Embedding

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DeepEarth Architecture

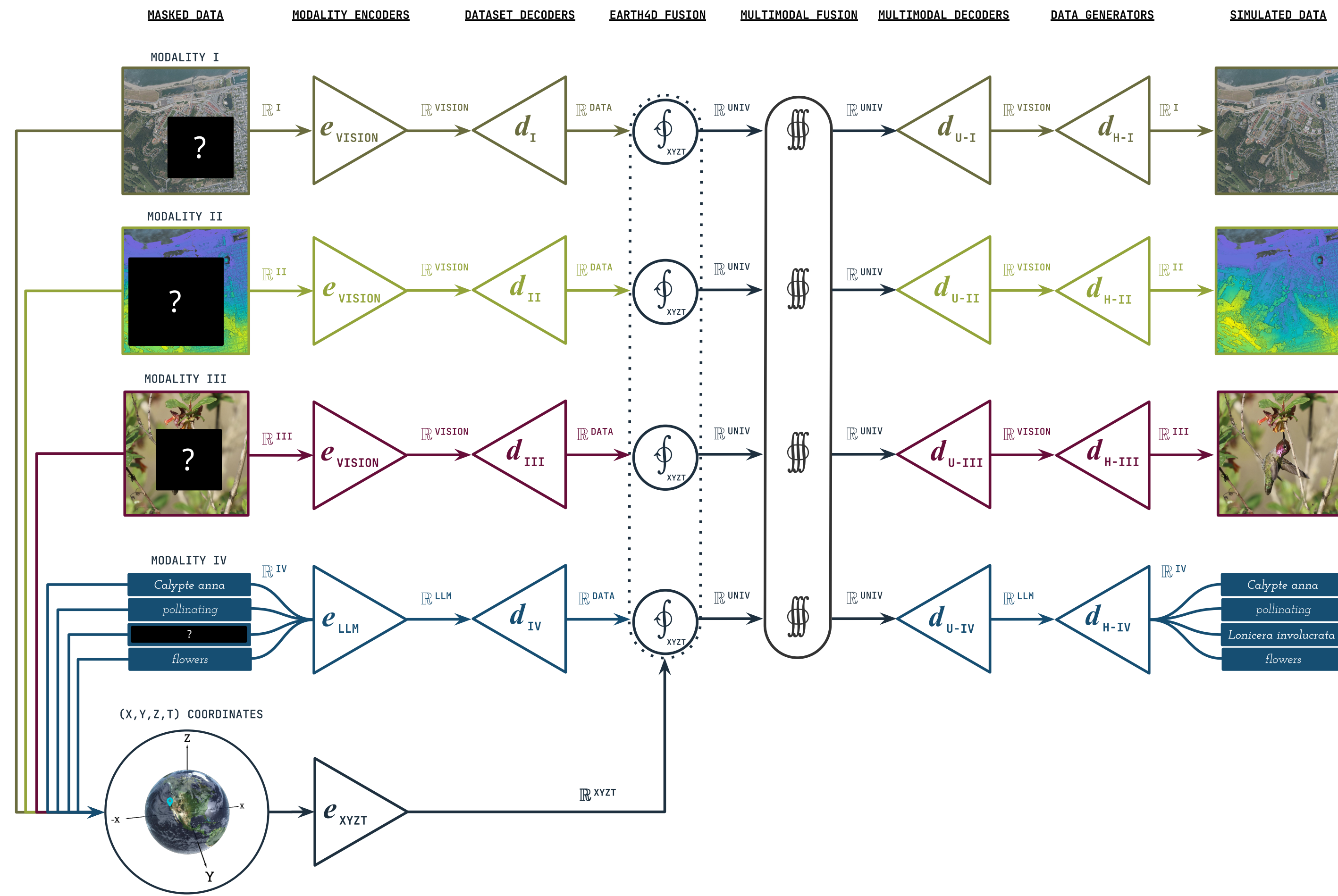


Figure 1. Self-supervised multi-modal world model for planetary science, simulation, & planning.

- Earth4D encodes (x, y, z, t) coordinates into learnable space-time embeddings
- DeepEarth fuses multi-modal (e.g. vision-language) and space-time embeddings
- Model trains by masked reconstruction of multi-modal data across space-time

Key Contributions

1. Space-Time Positional Encoder: Unify all kinds of physical data modalities
2. Multi-Scale 4D Geospatial Simulator: Bridge planetary-to-cellular dynamics
3. 4D Learned Hash Probing: Differentiably map $(x, y, z, t) \rightarrow$ embedding indices

Historical Exploratory Ecological Benchmark

Live Fuel Moisture Content (LFMC) measures vegetation water for wildfire risk. Training: Earth4D encodes (x, y, z, t) coordinates of LFMC metrics to 192D, concatenates with learnable plant species embeddings, MLP predicts LFMC. Historical result: A development run reported lower error than a pre-trained Vision Transformer with less input data. This comparison is hypothesis-generating, not confirmatory.

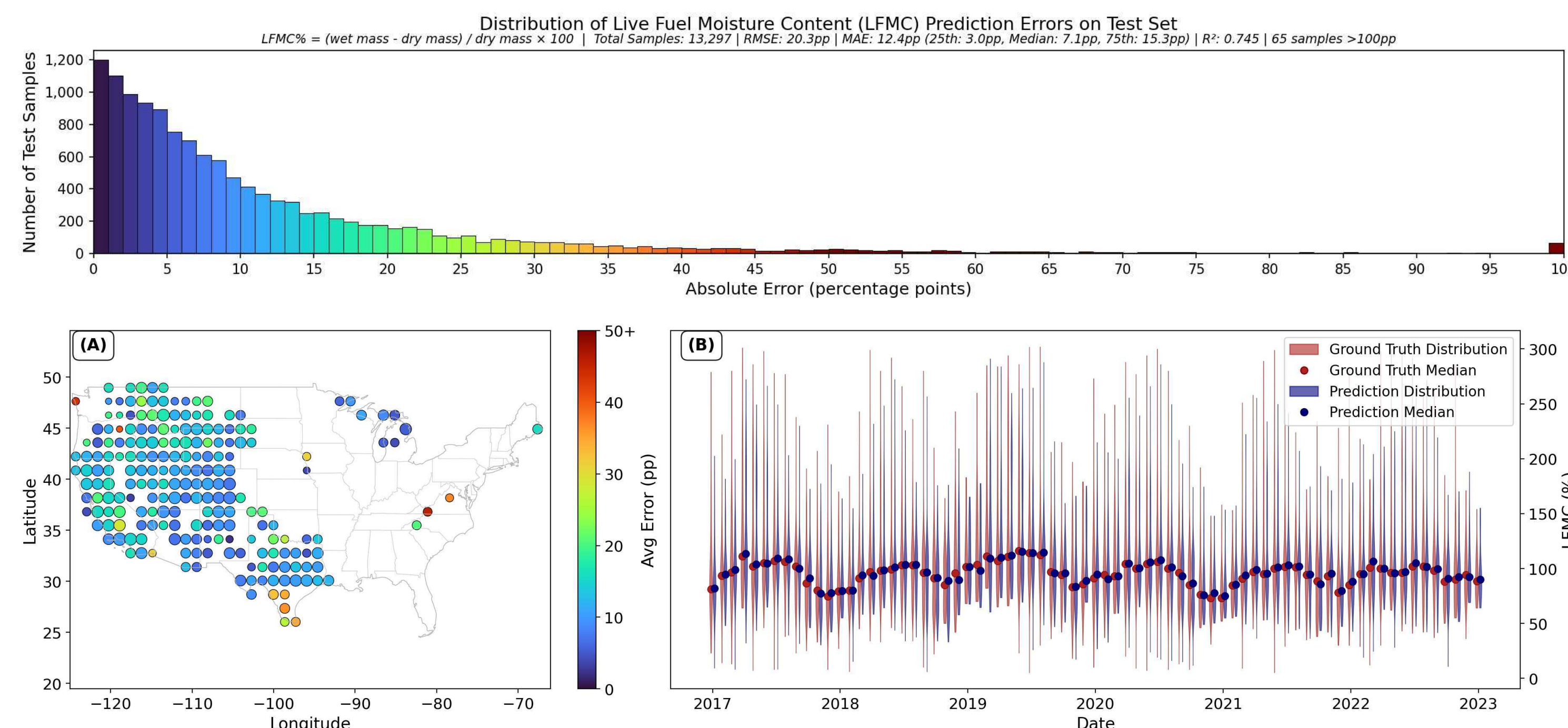


Figure 4. Earth4D LFMC test predictions on Allen Institute for AI (AI2) Globe-LFMC 2.0 benchmark.

Model	Data Inputs	MAE	RMSE	R ²
AI2 (Pre-trained ViT)	(x, y, z, t) + Species + Vision (Remote Sensing)	12.6pp	18.9pp	0.72
Earth4D	(x, y, z, t) + Species	11.7pp	18.7pp	0.783

Evidence status: These historical figures used test-set feedback during development and did not evaluate causal future rollout; they do not establish state of the art. Only coordinates and species names were used.

Earth4D Space-Time Positional Encoding

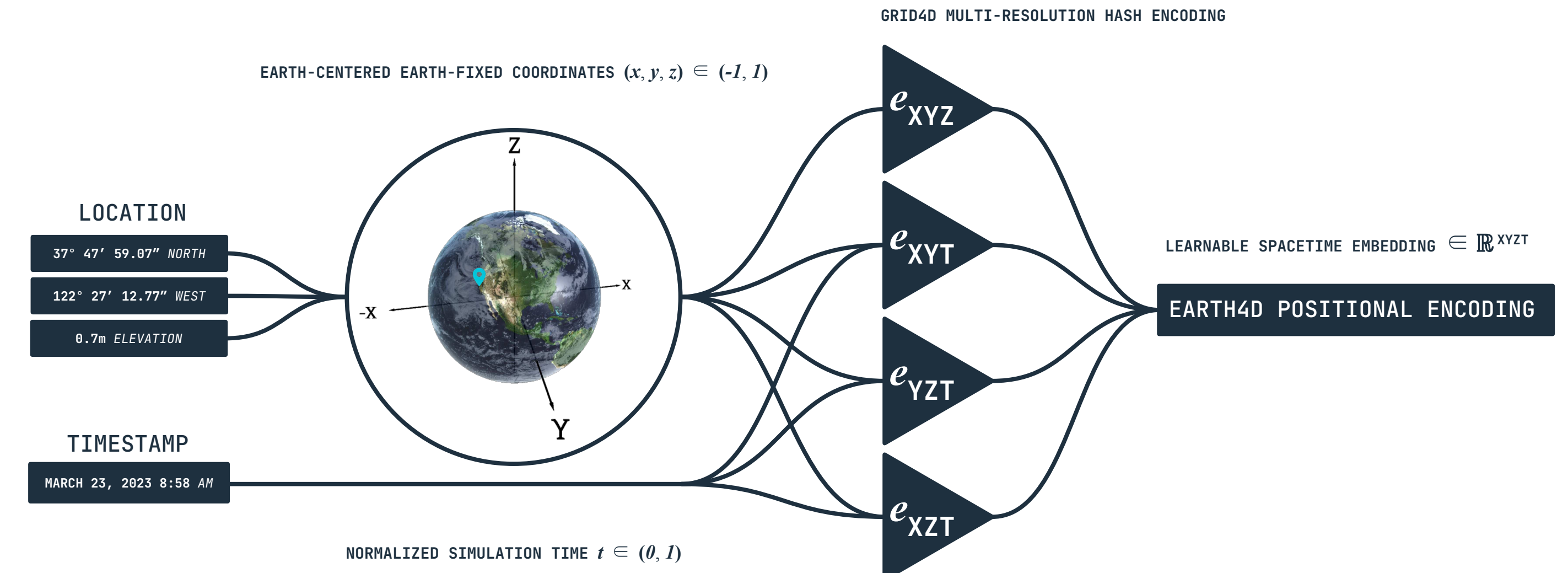


Figure 2. Multi-resolution hash encoding extended to 4D space-time across the planet over years.

Earth4D: multi-resolution hash encoder with four parallel 3D grids.
Spatial (xyz) : static structure. Spatio-temporal (xyt, yzt, xzt) : dynamics.
Geographic: Maps $(latitude, longitude, elevation, time) \rightarrow (x, y, z, t)$

Joint Embeddings Across Spatio-Temporal Scales

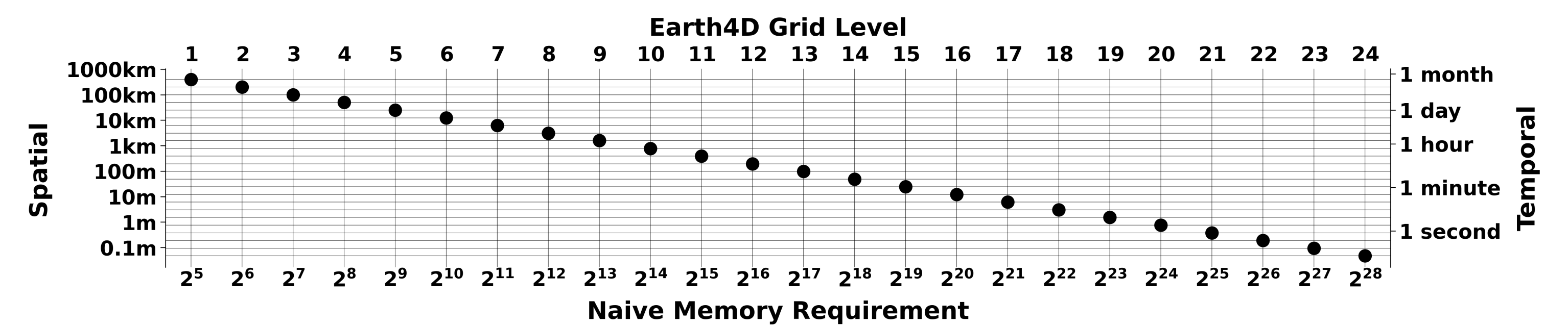


Figure 3. 24 resolution levels per grid (xyz, xyt, yzt, xzt) , up to 2^{22} entries each.
Output: 192D trainable embedding per (x, y, z, t) coordinate.

Learned Hash Probing

Hash encoding compresses features into fixed memory, but collisions hurt accuracy. Learned hash probing optimizes memory allocation end-to-end.

Historical exploratory figures: 33% fewer collisions; reported MAE 27% lower and R² 30% higher. Confirmation is pending.

Code Demo: Space-Time Positional Encoding

```
# https://github.com/legel/deepearth
from deepearth.encoders.xyzt.earth4d import Earth4D
world_model = Earth4D()
embeddings = world_model(
    # Bletchley Park (Turing breaks Enigma, 1941)
    (51.9976, -0.7416, 110, "1941-06-01 09:00 GMT"),
    # Carnegie Mellon (Hinton invents Boltzmann Machines, 1985)
    (40.4433, -79.9436, 270, "1985-01-15 10:00 ET"),
    # CERN (Berners-Lee invents WWW, 1989)
    (46.2330, 6.0557, 430, "1989-03-12 10:00 CET"),
    # Mila, Quebec (World Modeling Workshop 2026)
    (45.5308, -73.6128, 63, "2026-02-04 11:00 ET"),
)
# embeddings.shape: [4, 192] -- trainable space-time features
```

References

1. Müller et al. "Instant Neural Graphics Primitives with a Multiresolution Hash Encoding." *ACM SIGGRAPH*, 2022.
2. Takikawa et al. "Compact Neural Graphics Primitives with Learned Hash Probing." *SIGGRAPH Asia*, 2023.
3. Xu et al. "Grid4D: 4D Decomposed Hash Encoding for High-Fidelity Dynamic Gaussian Splatting." *NeurIPS*, 2024.
4. Yebra et al. "Globe-LFMC 2.0, an enhanced and updated dataset for live fuel moisture content research." *Scientific Data*, 2024.
5. Tseng et al. "Galileo: Learning Global and Local Features of Many Remote Sensing Modalities." *ICML*, 2025.
6. Johnson et al. "High-Resolution LFMC Maps for Wildfire Risk From Multimodal Earth Observation Data." *PMLR*, 2025.